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The prevalence and distribution of HPV infection and subtypes in oral squamous cell carcinoma in Africa: A systematic review protocol.

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Keywords:	Epidemiology < ONCOLOGY, Head & neck tumours < ONCOLOGY, ORAL MEDICINE

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Title:

The prevalence and distribution of HPV infection and subtypes in oral squamous cell carcinoma in Africa: A systematic review protocol.

Haly Holmes*
Department of Oral Medicine and Periodontology, University of The Western Cape Dental Faculty, Cape Town, South Africa

Amir Afrogeh
Department of Oral Pathology, University of the Western Cape and National Health Laboratory Service (NHLS)

Henry Adeola
Division of Dermatology, Department of Medicine, Faculty of Health Sciences and Groote Schuur Hospital, University of Cape Town, Cape Town South Africa.

Faheema Kimmie-Dhansay
Division of Research and Postgraduate Studies, Faculty of Dentistry, University of the Western Cape, Cape Town, South Africa

Mark Engel
Department of Medicine, Faculty of Health Sciences, University of Cape Town, Cape Town, South Africa

***Corresponding Author**
Haly Holmes, Private Bag X1, Tygerberg, Parow, 7505, hholmes@uwc.ac.za, + 27 21 9372287

KEYWORDS
"Head and Neck Neoplasms (MeSH), "Mouth Neoplasms" (MeSH, "HPV and Human papillomavirus", "Papillomaviradae" (MeSH), "Africa", prevalence.

WORD COUNT: 1841

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Abstract

Introduction

Human Papillomavirus is an established risk factor for oropharyngeal squamous cell carcinoma, regardless of a history of other known risk factors such as alcohol and tobacco. While cases of HPV-related OPSSC are increasing in the United States, Europe and South Central Asian countries, little is known about the impact of the disease on the African continent.

Methods and analysis:

We describe a protocol for a systematic review to synthesize the best current evidence to assess the disease burden in Africa. Electronic databases including EBSCOhost, MEDLINE, CINAHL, ACADEMIC SEARCH COMPLETE, ScienceDirect, Web of Science, Scopus, SciCENTRAL, Cochrane Library, PROSPERO) and WorldCAT will be comprehensively searched for studies reporting on the defined outcomes, in Africa, published from 1985 (when HPV was first reported) to the latest current entries, with no language restriction. Supplemental handsearching of grey literature, conference abstract proceedings, reference lists of included studies and citations in Google Scholar will be conducted. Authors will be contacted, where necessary, to assist with missing data. A customized data extraction form, with specified criteria, will be used for data extraction. Overall study quality assessment will be done using an appropriate risk of bias tool suited to the study design. Where available, qualitative data from studies reporting on the outcomes will be captured on the data extraction form. Using Stata software, we will apply the random-effects metanalysis model to aggregate prevalence estimates with 95% CI, incorporating the Freeman-Tukey transformation to account for between-study variability. A narrative report of the findings will be presented where data are insufficient in terms of the outcome/s. Subgroup analysis will be done subject to sufficient available data.

Ethics and dissemination:

Ethics approval or written consent is not required as the review will be conducted using published data. The findings will be distributed through a peer-review publication and conference presentation.

Patient and Public involvement

There is no patient nor public involvement in the study.

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Strengths and limitations of this study

- This study will present the best evidence on prevalence of HPV OSSC in Africa
- The results will have implications for policy, practice and further research and inform health and policy makers on current practices relative to accepted international standards
- This review will rely on reporting within studies and may not represent the true prevalence of HPV OSCC.
- The review findings may not be representative of the true population based HPV OSCC on the African continent

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INTRODUCTION

Human Papillomavirus (HPV) is an established risk factor for oral squamous cell carcinoma, particularly within the oropharyngeal region[1 2], regardless of a history of other known risk factors such as alcohol, tobacco or both. HPV Head and Neck Cancer (HNC) represents a distinct disease entity compared with HPV negative tumours. Patients with HPV + tumours present at a younger age and are less likely to partake in excessive smoking and alcohol use [2-4]. Although the annual incidence of smoking-related Head and Neck Squamous Cell Carcinoma's (HNSCC) has decreased significantly over the recent decade at a rate of 1,85% annually, HPV-associated HNSCC incidence has increased significantly at a rate 0.8% annually [3].

Context for this review

HPV belongs to a family of viruses that is sexually transmitted [5]. It is a double stranded DNA virus, whose genome contains open reading frames which are potential coding sites for 6 early (E) proteins and 2 late (L) proteins [6], implicated in oncogenic infection by HPV. Various diagnostic techniques to detect HPV infection in OSSC have been used in studies. In 2018, the American Society of Clinical Oncology recommended that p16 immunohistocytochemistry positivity be used as a surrogate marker for High risk HPV expression, with at least 70% nuclear and cytoplasmic expression [2 7].

Globally, the exact prevalence and distribution of HPV in OSSC specifically is uncertain and needs to be determined as its incidence in head and neck cancer reportedly varies by geographical location. Cases of HPV-related OSSC are reportedly increasing in the United States, Europe and South-Central Asian countries [4], but little is known about the burden and impact of the disease on the African continent [8]. In the African context, the proportion of patients with HPV-positive disease is hard to estimate, as many referral centres do not have access to routine testing. It has been hypothesised that the HPV pandemic will still emerge in developing countries owing to their burden of sexually transmitted infectious diseases [9].

Oral cancer is among the top ten cancers by incidence and deaths in low- or middle-income countries, as opposed to high income countries, with an annual global incidence of > 650 000 and > 330 000 deaths cases. Head and neck squamous cell carcinoma (HNSCC) are the sixth

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most common cancer worldwide, approximately 50% of which occurs within the oral cavity. While the National Cancer Institute reports a marginal global decrease in the incidence of head and neck cancers, whose primary risk factors include smoking and tobacco, Human Papillomavirus (HPV) associated Head and Neck Cancers (HNC's) have increased considerably since the debut isolation/discovery of HPV in a subset of HNC's (particularly oropharyngeal carcinomas) [10].

The majority (90%) of HNC's arise from the squamous epithelium lining of the oral cavity, larynx, pharynx and rarely nasal cavity [8 11]. The oral cavity and oropharynx are separate regions that border each other but do not overlap [12]. Oral SSC arise from the lips tongue (anterior two thirds of the tongue -anterior to circumvallate papillae), floor of mouth and hard palate and retromolar trigone oral cavity. The oral cavity anatomic sites include labial mucosa, buccal mucosa, floor of mouth, alveolar ridge and gingiva, anterior two thirds of the tongue (anterior to circumvallate papillae). The oropharynx consists of the soft palate, base (posterior third of tongue), (palatine tonsil, palatoglossal folds, valleculae and posterior pharyngeal wall) and back of throat. These distinct anatomic borders separate the two sites from above the, the junction of the hard and soft palate, and from below, the circumvallate papillae [2 12]. We propose to conduct a systematic review for the best evidence on the prevalence of HPV OSSC in adults in Africa.

Objectives:

This protocol describes our intended systematic review to identify, appraise and synthesize best evidence to address the question: What is the prevalence and distribution of HPV subtypes in OSSC in Africa? The specific objectives/secondary outcomes are 1) to determine the distribution of HPV associated OPSSC in Africa (by geographical site, oral cavity distribution, HPV subtype, demographic profile age and gender); 2) To describe the histological phenotype of HPV associated OSSC in Sub Saharan Africa; 3) to describe the molecular phenotype/signatures of HPV associated OSSC such as aneuploidy, TRAF3 loss, PIK3CA activating mutations, E2F1 amplifications); 4) to describe the methods used to profile the HPV subtypes

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METHODS AND DESIGN:

The protocol for a systematic review on the prevalence of HPV OSSC is detailed in this article and has been submitted for registration in PROSPERO. We will utitlize specified inclusion and exclusion criteria to select studies for this systematic review, which will be reported according to PRISMA guidelines (Preferred Reporting Items for Systematic reviews Protocols) [13]. We will conduct a comprehensive literature search using the proposed search strategy and databases.

Study eligibility criteria

Inclusion criteria

Study design

We will include will include observational studies such as cross sectional, case control and case series studies.

Population/participants

Participants will be all adults, older or equal to 18 years of age with OSSC, living in AFRICA, diagnosed from Fresh Frozen Paraffin embedded (FFPE) specimens, confirmed by a pathologist and whose HPV status is confirmed by a standardized diagnostic test.

Outcome:

The prevalence of HPV and subtypes in OSSC in Africa.

Types of outcome measures:

1. To determine HPV prevalence by:
    a. Geographical distribution
    b. Oral distribution
    c. HPV subtype
    d. Demographic profile (age and gender)
2. To determine the histological phenotype of HPV associated OSSC in Sub Saharan Africa

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3. To describe the molecular phenotype/ signature of HPV associated OSSC such as aneuploidy, TRAF3 loss, PIK3CA activating mutations, E2F1amplifications.
4. Describe the methods used to profile HPV subtypes.

Ex*clusion criteria:*

We will exclude articles where full texts could not be obtained from the search, as were case reports, case series, qualitative measures/aspects and studies reporting children with HPV OPSCC.

Search strategy for identification of studies

We will use the CoCoPop framework (condition, context, population) for our search strategy. Studies must meet the inclusion criteria and report data pertaining to the numerator (Number of HPV) and denominator (number of people with OSSC), which we will extract.

All relevant studies in Africa published from 1985 until date of searching, will be comprehensively searched and retrieved from the following databases: EBSCO Host, MEDLINE, ACADEMIC SEARCH COMPLETE, ScienceDirect, ERIC, WEB OF SCIENCE, SciELO), EMBASE, CENTRAL, COCHRANE LIBRARY,PROSPERO, SCOPUS).
Grey literature databases and other literature sources including conference abstract proceedings and NETD and WORLD CAT thesis databases will be searched. In addition, searches will be supplemented by hand searching reference lists of included studies and citations in Google Scholar, to identify other potential studies for consideration. Online registers will be searched for unpublished and ongoing studies. Study authors will be contacted where full text cannot be accessed or for any missing information.

A wide search strategy will be used to reduce publication bias, with no language restrictions. We will use varying combinations of text words and MeSH (Medical subject headings) terms used in PubMed/Medline to develop a preliminary search (table 1) and adapt the search strategy for every other database. Keywords proposed will be “Head and Neck Neoplasms (MeSH), “Mouth Neoplasms” (MeSH)”, “HPV and Human papillomavirus", “Papillomaviradae” (MeSH), “Africa”, prevalence. We will use the African countries filter was used [14], to ensure inclusion of all African countries in the search.

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Data collection and analysis

Two reviewers will independently screen Titles and abstracts identified through the literature searches for eligibility. Two authors will then access the full text articles of the potential studies screened to compare the findings. Discrepancies will be resolved by discussion and consensus. Reasons for exclusion of any studies will be provided.

Data Extraction and Management

Search results will be managed using Rayyan systematic review web application [15]. For each included article, two reviewers will independently extract descriptive and outcome data, using a data extraction form designed/customized for that purpose. Any discrepancies will be resolved by discussion and consensus. Disagreements will be resolved by consensus or in consultation with a third reviewer. The data capturing sheet will include study characteristics such as author/s, country name , year of publication, national population, region (rural or urban), age range, sex, smoking status, study design, criteria for sample selection as well as sample material (paraffin- embedded) sample size, method of ascertainment of OSCC and HPV status, HPV subtype, diagnostic criteria used. The final data from included studies will be managed and analysed with Stata software. Prevalence figures and 95% confidence intervals (CIs) will be extracted or calculated from the available data. We will deal with missing data by contacting the corresponding authors.

Risk of bias assessment for included studies

Relevant risk of bias tools will be used according to the study design [16](table 2). Two independent reviewers will assess the methodological quality of the included literature for internal and external validity and general applicability of results. The adapted version [17]of this 10-item rating system will be used on domains specified within the inclusion criteria and outcomes (such as sampling frame and size, methods used to confirm diagnosis, outcomes and statistical reporting). Each item will be assigned a score of 1(yes) and 0 (no). The sum of all the score items from 0-10 will generate an overall quality score. Studies with scores of 8 or more, between 5-6 and less than 5, will be considered as low, medium and high risk respectively (table 3).

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Strategy for Data Synthesis and Analysis

Where available, quantitative data from studies measuring and reporting on the outcomes will be captured on a data extraction form and a summary of the literature will be presented in data tables. Using Stata software, we will apply the random-effects metanalysis model to aggregate prevalence estimates with 95% CI, incorporating the Freeman-Tukey transformation to account for between-study variability.[18]

A narrative report of the results will be done where there are not enough studies measuring or reporting on the outcome/s. Subgroup analysis will be performed and stratified by secondary outcomes, provided there are sufficient data.

Presenting and reporting of results

Search results will be presented in the form of flow diagrams, to show the study selection and inclusion process. We will elaborate on the reason for excluded studies and provide a narrative description of outcomes for which there is insufficient data available for the desired outcome measures. Outcomes will be represented by Forest plots and tables where possible.

Patient and public involvement

There is no patient nor public involvement in the study.

DISCUSSION

Anticipated significance of the study

This systematic review will identify and present the available evidence for the prevalence of HPV OSSC on the African continent. The findings of this systematic review will have implications for policy, practice and research. It will inform /provide the status quo on burden of the disease burden on the African continent and diagnostic methods used for detection. The relevance of the findings on HPV prevalence on the African continent will inform health and policy makers of the current practice relative to accepted international standards. We will discuss strengths and limitations of the study, highlighting any evidence gaps on the literature.

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The findings of this systematic review will be compared with that of other global systematic reviews, to compare the clinical picture of HPV OSSC in Africa.

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Authors Contributions

HH conceptualized the review. HH and ME developed the design for the protocol and will be involved in data acquisition. HH and ME will analyze the data and HH, ME, AA, HA AND FK will be involved in the interpretation of the results. All authors have consented to publication.

Funding:

This research received no specific grant from any finding agency in the public, commercial or non-profit sectors.

Competing Interest

None declared

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Table 1: Search Strategy used in MEDLINE : MeSH (Medical subject headings) and Keywords
("africa"[MeSH Terms] OR "africa"[All Fields] OR "africa s"[All Fields] OR "africas"[All Fields] OR "Comoros"[All Fields] OR "Djibouti"[All Fields] OR "Madagascar"[All Fields] OR "Malawi"[All Fields] OR "Seychelles"[All Fields] OR "Cameroon"[All Fields] OR "Central African Republic"[All Fields] OR "Chad"[All Fields] OR "Equatorial Guinea"[All Fields] OR ("gabon"[MeSH Terms] OR "gabon"[All Fields]) OR "Morocco"[All Fields] OR "South Sudan"[All Fields] OR "Sudan"[All Fields] OR "Botswana"[All Fields] OR "Lesotho"[All Fields] OR "Swaziland"[All Fields] OR ("eswatini"[MeSH Terms] OR "eswatini"[All Fields]) OR "Benin"[All Fields] OR "Burkina Faso"[All Fields] OR "Cape Verde"[All Fields] OR "Ghana"[All Fields] OR "Guinea"[All Fields] OR "Guinea-Bissau"[All Fields] OR "Mauritania"[All Fields] OR "Niger"[All Fields] OR "Senegal"[All Fields] OR "Sierra Leone"[All Fields] OR "Togo"[All Fields] OR ("burundi"[MeSH Terms] OR "burundi"[All Fields]) OR ("eritrea"[MeSH Terms] OR "eritrea"[All Fields]) OR ("ethiopia"[MeSH Terms] OR "ethiopia"[All Fields] OR "ethiopia s"[All Fields]) OR ("kenya"[MeSH Terms] OR "kenya"[All Fields] OR "kenya s"[All Fields]) OR ("mozambique"[MeSH Terms] OR "mozambique"[All Fields] OR "mozambique s"[All Fields]) OR ("rwanda"[MeSH Terms] OR "rwanda"[All Fields] OR "rwanda s"[All Fields]) OR ("somalia"[MeSH Terms] OR "somalia"[All Fields]) OR ("tanzania"[MeSH Terms] OR "tanzania"[All Fields] OR "tanzania s"[All Fields]) OR ("uganda"[MeSH Terms] OR "uganda"[All Fields] OR "uganda s"[All Fields]) OR ("zambia"[MeSH Terms] OR "zambia"[All Fields] OR "zambia s"[All Fields]) OR ("zimbabwe"[MeSH Terms] OR "zimbabwe"[All Fields] OR "zimbabwe s"[All Fields]) OR ("angola"[MeSH Terms] OR "angola"[All Fields] OR "angola s"[All Fields]) OR ("algeria"[MeSH Terms] OR "algeria"[All Fields]) OR ("egypt"[MeSH Terms] OR "egypt"[All Fields] OR "egypt s"[All Fields]) OR ("tunisia"[MeSH Terms] OR "tunisia"[All Fields]) OR ("namibia"[MeSH Terms] OR "namibia"[All Fields]) OR "South Africa"[All Fields] OR "the gambia"[All Fields] OR ("liberia"[MeSH Terms] OR "liberia"[All Fields] OR "liberia s"[All Fields]) OR ("mali"[MeSH Terms] OR "mali"[All Fields]) OR "Democratic Republic of Congo"[All Fields] OR "Ivory coast"[All Fields] OR ("libya"[MeSH Terms] OR "libya"[All Fields]) OR

("mauritius"[MeSH Terms] OR "mauritius"[All Fields]) OR ("nigeria"[MeSH Terms] OR "nigeria"[All Fields] OR "nigeria s"[All Fields]) OR "Sao Tome"[All Fields] OR "sub africa*"[All Fields]) AND ("Papillomaviridae"[MeSH Terms] OR ("HPV"[Title/Abstract] OR "HUMAN PAPILLOMA VIRUS"[Title/Abstract] OR "HUMAN PAPILLOMAVIRUS"[Title/Abstract])) AND "Head and Neck Neoplasms"[MeSH Terms] AND ("PREVALENCE"[Title/Abstract] OR "INCIDENCE"[Title/Abstract] OR "PRESENCE"[Title/Abstract] OR "EPIDEMIOLOGY"[Title/Abstract])

Table 2: Quality assessment tool[16]

External Validity		**points**
1	Was the study's target population a close representation of the national population in relation to relevant variables?	
2.	Was the sampling frame a true/close representation of the target population?	
3	Was some form of random selection used to select the sample?	
4	Was the likelihood of non-response bias minimal	
Total Points		
Internal Validity		
5	Was data collected directly from patients as opposed to proxy?	
6	Was an acceptable case definition used in the study?	
7	Was the study instrument that measured the parameter of interest shown to have validity and reliability?	

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8	Was the same mode of data collection used for all participants?	
9	Was the length of the shortest prevalence period for the parameter of interest appropriate?	
10	Were the numerators and denominators for the parameters of interest appropriate?	
Total points		

Table 3: Key for measurement of risk of bias[17]:

Low (0-5 points)

Further research is very unlikely to change our confidence in the estimate

Medium (6-8 points)

Further research is likely to have an important impact on our confidence in the estimate and may change the estimate

High (> 8 points)

Further research is likely

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PRISMA-P 2015 Checklist

This checklist has been adapted for use with protocol submissions to *Systematic Reviews* from Table 3 in Moher D et al: Preferred reporting items for systematic review and meta-analysis protocols (PRISMA-P) 2015 statement. *Systematic Reviews* 2015 **4**:1

Section/topic	#	Checklist item	Information reported		Line number(s)
			Yes	No	
ADMINISTRATIVE INFORMATION					
Title					
Identification	1a	Identify the report as a protocol of a systematic review	☐		3
Update	1b	If the protocol is for an update of a previous systematic review, identify as such		☐	
Registration	2	If registered, provide the name of the registry (e.g., PROSPERO) and registration number in the Abstract	☐	☐	
Authors					
Contact	3a	Provide name, institutional affiliation, and e-mail address of all protocol authors; provide physical mailing address of corresponding author	☐		5
Contributions	3b	Describe contributions of protocol authors and identify the guarantor of the review	☐		286
Amendments	4	If the protocol represents an amendment of a previously completed or published protocol, identify as such and list changes; otherwise, state plan for documenting important protocol amendments		☐	
Support					
Sources	5a	Indicate sources of financial or other support for the review	☐		292
Sponsor	5b	Provide name for the review funder and/or sponsor		☐	
Role of sponsor/funder	5c	Describe roles of funder(s), sponsor(s), and/or institution(s), if any, in developing the protocol	☐		282
INTRODUCTION					
Rationale	6	Describe the rationale for the review in the context of what is already known	☐		100
Objectives	7	Provide an explicit statement of the question(s) the review will address with reference to participants, interventions, comparators, and outcomes (PICO)	☐		136
METHODS					

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Section/topic	#	Checklist item	Information reported		Line number(s)
			Yes	No	
Eligibility criteria	8	Specify the study characteristics (e.g., PICO, study design, setting, time frame) and report characteristics (e.g., years considered, language, publication status) to be used as criteria for eligibility for the review	☐		164,195
Information sources	9	Describe all intended information sources (e.g., electronic databases, contact with study authors, trial registers, or other grey literature sources) with planned dates of coverage	☐		196
Search strategy	10	Present draft of search strategy to be used for at least one electronic database, including planned limits, such that it could be repeated	☐		301
STUDY RECORDS					
Data management	11a	Describe the mechanism(s) that will be used to manage records and data throughout the review	☐		222
Selection process	11b	State the process that will be used for selecting studies (e.g., two independent reviewers) through each phase of the review (i.e., screening, eligibility, and inclusion in meta-analysis)	☐		215
Data collection process	11c	Describe planned method of extracting data from reports (e.g., piloting forms, done independently, in duplicate), any processes for obtaining and confirming data from investigators	☐		216, 232
Data items	12	List and define all variables for which data will be sought (e.g., PICO items, funding sources), any pre-planned data assumptions and simplifications	☐		191
Outcomes and prioritization	13	List and define all outcomes for which data will be sought, including prioritization of main and additional outcomes, with rationale	☐		169-181
Risk of bias in individual studies	14	Describe anticipated methods for assessing risk of bias of individual studies, including whether this will be done at the outcome or study level, or both; state how this information will be used in data synthesis	☐		238
DATA					
Synthesis	15a	Describe criteria under which study data will be quantitatively synthesized	☐		250
	15b	If data are appropriate for quantitative synthesis, describe planned summary measures, methods of handling data, and methods of combining data from studies, including any planned exploration of consistency (e.g., I^2, Kendall's tau)	☐		252
	15c	Describe any proposed additional analyses (e.g., sensitivity or subgroup analyses, meta-regression)	☐		257
	15d	If quantitative synthesis is not appropriate, describe the type of summary planned	☐		256
Meta-bias(es)	16	Specify any planned assessment of meta-bias(es) (e.g., publication bias across studies, selective reporting within studies)		☐	
Confidence in cumulative evidence	17	Describe how the strength of the body of evidence will be assessed (e.g., GRADE)		☐	

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The prevalence and distribution of HPV infection and subtypes in oral squamous cell carcinoma in Africa: A systematic review protocol.

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Primary Subject Heading:	Dentistry and oral medicine
Secondary Subject Heading:	Oncology, Epidemiology
Keywords:	Epidemiology < ONCOLOGY, Head & neck tumours < ONCOLOGY, ORAL MEDICINE

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Title:

The prevalence and distribution of HPV infection and subtypes in oral squamous cell carcinoma in Africa: A systematic review protocol.

Haly Holmes*
Department of Oral Medicine and Periodontology, University of The Western Cape Dental Faculty, Cape Town, South Africa

Amir Afrogeh
Department of Oral Pathology, University of the Western Cape and National Health Laboratory Service (NHLS)

Henry Adeola
Division of Dermatology, Department of Medicine, Faculty of Health Sciences and Groote Schuur Hospital, University of Cape Town, Cape Town South Africa.

Faheema Kimmie-Dhansay
Division of Research and Postgraduate Studies, Faculty of Dentistry, University of the Western Cape, Cape Town, South Africa

Mark Engel
Department of Medicine, Faculty of Health Sciences, University of Cape Town, Cape Town, South Africa

***Corresponding Author**
Haly Holmes, Private Bag X1, Tygerberg, Parow, 7505, hholmes@uwc.ac.za, + 27 21 9372287

KEYWORDS
"Head and Neck Neoplasms (MeSH), "Mouth Neoplasms" (MeSH, "HPV and Human papillomavirus", "Papillomaviradae" (MeSH), "Africa", prevalence.

WORD COUNT: 1881

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Abstract

Introduction

Human Papillomavirus is an established risk factor for oropharyngeal squamous cell carcinoma, regardless of a history of other known risk factors such as alcohol and tobacco. While cases of HPV-related OPSSC are increasing in the United States, Europe and South Central Asian countries, little is known about the impact of the disease on the African continent.

Methods and analysis:

We describe a protocol for a systematic review to synthesize the best current evidence to assess the disease burden in Africa. Electronic databases including EBSCOhost, MEDLINE, CINAHL, ACADEMIC SEARCH COMPLETE, ScienceDirect, Web of Science, Scopus, SciCENTRAL, Cochrane Library, PROSPERO) and WorldCAT will be comprehensively searched for studies reporting on the defined outcomes, in Africa, published from 1985 (when HPV was first reported) to the latest current entries, with no language restriction. Supplemental handsearching of grey literature, conference abstract proceedings, reference lists of included studies and citations in Google Scholar will be conducted. Authors will be contacted, where necessary, to assist with missing data. A customized data extraction form, with specified criteria, will be used for data extraction. Overall study quality assessment will be done using an appropriate risk of bias tool suited to the study design. Where available, qualitative data from studies reporting on the outcomes will be captured on the data extraction form. Using Stata software, we will apply the random-effects metanalysis model to aggregate prevalence estimates with 95% CI, incorporating the Freeman-Tukey transformation to account for between-study variability. A narrative report of the findings will be presented where data are insufficient in terms of the outcome/s. Subgroup analysis will be done subject to sufficient available data.

Ethics and dissemination:

Ethics approval or written consent is not required as the review will be conducted using published data. The findings will be distributed through a peer-review publication and conference presentation.

Patient and Public involvement

There is no patient nor public involvement in the study.

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Strengths and limitations of this study

- This study will present the best evidence on prevalence of HPV OSSC in Africa
- The results will have implications for policy, practice and further research and inform health and policy makers on current practices relative to accepted international standards
- This review will rely on reporting within studies and may not represent the true prevalence of HPV OSCC.
- The review findings may not be representative of the true population-based HPV OSCC on the African continent

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INTRODUCTION

Human Papillomavirus (HPV) is an established risk factor for oral squamous cell carcinoma, particularly within the oropharyngeal region[1,2], regardless of a history of other known risk factors such as alcohol, tobacco or both. HPV Head and Neck Cancer (HNC) represents a distinct disease entity compared with HPV negative tumours. Patients with HPV + tumours present at a younger age and are less likely to partake in excessive smoking and alcohol use [2-4]. Although the annual incidence of smoking-related Head and Neck Squamous Cell Carcinoma's (HNSCC) has decreased significantly over the recent decade at a rate of 1,85% annually, HPV-associated HNSCC incidence has increased significantly at a rate 0.8% annually [3].

Context for this review

HPV belongs to a family of viruses that is sexually transmitted [5]. It is a double stranded DNA virus, whose genome contains open reading frames which are potential coding sites for 6 early (E) proteins and 2 late (L) proteins [6], implicated in oncogenic infection by HPV. Various diagnostic techniques to detect HPV infection in OSSC have been used in studies. In 2018, the American Society of Clinical Oncology recommended that p16 immunohistocytochemistry positivity be used as a surrogate marker for High risk HPV expression, with at least 70% nuclear and cytoplasmic expression [2 7].

Globally, the exact prevalence and distribution of HPV in OSSC specifically is uncertain and needs to be determined as its incidence in head and neck cancer reportedly varies by geographical location. Cases of HPV-related OSSC are reportedly increasing in the United States, Europe and South-Central Asian countries [4], but little is known about the burden and impact of the disease on the African continent [8]. In the African context, the proportion of patients with HPV-positive disease is hard to estimate, as many referral centres do not have access to routine testing. It has been hypothesised that the HPV pandemic will still emerge in developing countries owing to their burden of sexually transmitted infectious diseases [9].

Oral cancer is among the top ten cancers by incidence and deaths in low- or middle-income countries, as opposed to high income countries, with an annual global incidence of > 650 000 and > 330 000 deaths cases. Head and neck squamous cell carcinoma (HNSCC) are the sixth

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most common cancer worldwide, approximately 50% of which occurs within the oral cavity. While the National Cancer Institute reports a marginal global decrease in the incidence of head and neck cancers, whose primary risk factors include smoking and tobacco, Human Papillomavirus (HPV) associated Head and Neck Cancers (HNC's) have increased considerably since the debut isolation/discovery of HPV in a subset of HNC's (particularly oropharyngeal carcinomas) [10].

The majority (90%) of HNC's arise from the squamous epithelium lining of the oral cavity, larynx, pharynx and rarely nasal cavity [8 11]. The oral cavity and oropharynx are separate regions that border each other but do not overlap [12]. Oral SSC arise from the lips tongue (anterior two thirds of the tongue -anterior to circumvallate papillae), floor of mouth and hard palate and retromolar trigone oral cavity. The oral cavity anatomic sites include labial mucosa, buccal mucosa, floor of mouth, alveolar ridge and gingiva, anterior two thirds of the tongue (anterior to circumvallate papillae). The oropharynx consists of the soft palate, base (posterior third of tongue), (palatine tonsil, palatoglossal folds, valleculae and posterior pharyngeal wall) and back of throat. These distinct anatomic borders separate the two sites from above the, the junction of the hard and soft palate, and from below, the circumvallate papillae [2 12]. We propose to conduct a systematic review for the best evidence on the prevalence of HPV OSSC in adults in Africa.

Objectives:

This protocol describes our intended systematic review to identify, appraise and synthesize best evidence to address the question: What is the prevalence and distribution of HPV subtypes in OSSC in Africa? The specific objectives/secondary outcomes are 1) to determine the distribution of HPV associated OPSSC in Africa (by geographical site, oral cavity distribution, HPV subtype, demographic profile age and gender); 2) To describe the histological phenotype of HPV associated OSSC in Sub Saharan Africa; 3) to describe the molecular phenotype/signatures of HPV associated OSSC such as aneuploidy, TRAF3 loss, PIK3CA activating mutations, E2F1 amplifications); 4) to describe the methods used to profile the HPV subtypes

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METHODS AND DESIGN:

The protocol for a systematic review on the prevalence of HPV OSSC is detailed in this article and has been submitted for registration in PROSPERO. We will utilize specified inclusion and exclusion criteria to select studies for this systematic review, which will be reported according to PRISMA guidelines (Preferred Reporting Items for Systematic reviews Protocols) [13]. We will conduct a comprehensive literature search using the proposed search strategy and databases.

Study eligibility criteria

Inclusion criteria

Study design

We will include will include observational studies such as cross sectional, case control and case series studies.

Population/participants

Participants will be all adults, older or equal to 18 years of age with OSSC, living in AFRICA, diagnosed from formalin-fixed paraffin-embedded (FFPE) specimens, confirmed by a pathologist and whose HPV status is confirmed by a standardized diagnostic test.

Outcome:

The prevalence of HPV and subtypes in OSSC in Africa.

Types of outcome measures:

1. To determine HPV prevalence by:
    a. Geographical distribution
    b. Oral distribution
    c. HPV subtype
    d. Demographic profile (age and gender)
2. To determine the histological phenotype of HPV associated OSSC in Sub Saharan Africa

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3. To describe the molecular phenotype/ signature of HPV associated OSSC such as aneuploidy, TRAF3 loss, PIK3CA activating mutations, E2F1amplifications.
4. Describe the methods used to profile HPV subtypes.

Ex*clusion criteria:*

We will exclude articles where full texts could not be obtained from the search, as were case reports, case series, qualitative measures/aspects and studies reporting children with HPV OPSCC.

Search strategy for identification of studies

We will use the CoCoPop framework (condition, context, population) for our search strategy. Studies must meet the inclusion criteria and report data pertaining to the numerator (Number of HPV) and denominator (number of people with OSSC), which we will extract.

All relevant studies in Africa published from 1985 until date of searching, will be comprehensively searched and retrieved from the following databases: EBSCO Host, MEDLINE, ACADEMIC SEARCH COMPLETE, ScienceDirect, ERIC, WEB OF SCIENCE, SciELO), EMBASE, CENTRAL, COCHRANE LIBRARY, PROSPERO, SCOPUS).
Grey literature databases and other literature sources including conference abstract proceedings and NETD and WORLD CAT thesis databases will be searched. In addition, searches will be supplemented by hand searching reference lists of included studies and citations in Google Scholar, to identify other potential studies for consideration. Online registers will be searched for unpublished and ongoing studies. Study authors will be contacted where full text cannot be accessed or for any missing information.

A wide search strategy will be used to reduce publication bias, with no language restrictions. We will use varying combinations of text words and MeSH (Medical subject headings) terms used in PubMed/Medline to develop a preliminary search (table 1) and adapt the search strategy for every other database. Keywords proposed will be “Head and Neck Neoplasms (MeSH), “Mouth Neoplasms” (MeSH)”, “HPV and Human papillomavirus", “Papillomaviradae” (MeSH), “Africa”, prevalence. We will use the African countries filter was used [14], to ensure inclusion of all African countries in the search.

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Data collection and analysis

Two reviewers will independently screen Titles and abstracts identified through the literature searches for eligibility. Two authors will then access the full text articles of the potential studies screened to compare the findings. Discrepancies will be resolved by discussion and consensus. Reasons for exclusion of any studies will be provided.

Data Extraction and Management

Search results will be managed using Rayyan systematic review web application [15]. For each included article, two reviewers will independently extract descriptive and outcome data, using a data extraction form designed/customized for that purpose. Any discrepancies will be resolved by discussion and consensus. Disagreements will be resolved by consensus or in consultation with a third reviewer. The data capturing sheet will include study characteristics such as author/s, country name, year of publication, national population, region (rural or urban), age range, sex, smoking status, study design, criteria for sample selection as well as sample material (paraffin- embedded) sample size, method of ascertainment of OSCC and HPV status, HPV subtype, diagnostic criteria used. The final data from included studies will be managed and analysed with Stata software. Prevalence figures and 95% confidence intervals (CIs) will be extracted or calculated from the available data. We will deal with missing data by contacting the corresponding authors.

Risk of bias assessment for included studies

Relevant risk of bias tools will be used according to the study design [16](table 2). Two independent reviewers will assess the methodological quality of the included literature for internal and external validity and general applicability of results. The adapted version [17]of this 10-item rating system will be used on domains specified within the inclusion criteria and outcomes (such as sampling frame and size, methods used to confirm diagnosis, outcomes and statistical reporting). Each item will be assigned a score of 1(yes) and 0 (no). The sum of all the score items from 0-10 will generate an overall quality score. Studies with scores of 8 or more, between 5-6 and less than 5, will be considered as low, medium and high risk respectively (table 3).

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Strategy for Data Synthesis and Analysis

Where available, quantitative data from studies measuring and reporting on the outcomes will be captured on a data extraction form and a summary of the literature will be presented in data tables. Using Stata software, we will apply the random-effects metanalysis model to aggregate prevalence estimates with 95% CI, incorporating the Freeman-Tukey transformation to account for between-study variability.[18]

A narrative report of the results will be done where there are not enough studies measuring or reporting on the outcome/s. Subgroup analysis will be performed and stratified by secondary outcomes, provided there are sufficient data.

Presenting and reporting of results

Search results will be presented in the form of flow diagrams, to show the study selection and inclusion process. We will elaborate on the reason for excluded studies and provide a narrative description of outcomes for which there is insufficient data available for the desired outcome measures. Outcomes will be represented by Forest plots and tables where possible.

Patient and public involvement

There is no patient nor public involvement in the study.

DISCUSSION

Anticipated significance of the study

This systematic review will identify and present the available evidence for the prevalence of HPV OSSC on the African continent. The findings of this systematic review will have implications for policy, practice and research. It will inform /provide the status quo on burden of the disease burden on the African continent and diagnostic methods used for detection. The relevance of the findings on HPV prevalence on the African continent will inform health and policy makers of the current practice relative to accepted international standards. We will discuss strengths and limitations of the study, highlighting any evidence gaps on the literature.

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The findings of this systematic review will be compared with that of other global systematic reviews, to compare the clinical picture of HPV OSSC in Africa.

Acknowledgements

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Authors Contributions

HH conceptualized the review. HH and ME developed the design for the protocol and will be involved in data acquisition. HH and ME will analyze the data and HH, ME, AA, HA AND FK will be involved in the interpretation of the results. All authors have consented to publication.

Funding:

This research received no specific grant from any finding agency in the public, commercial or non-profit sectors.

Competing Interest

None declared

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Table 1: Search Strategy used in MEDLINE : MeSH (Medical subject headings) and Keywords
("africa"[MeSH Terms] OR "africa"[All Fields] OR "africa s"[All Fields] OR "africas"[All Fields] OR "Comoros"[All Fields] OR "Djibouti"[All Fields] OR "Madagascar"[All Fields] OR "Malawi"[All Fields] OR "Seychelles"[All Fields] OR "Cameroon"[All Fields] OR "Central African Republic"[All Fields] OR "Chad"[All Fields] OR "Equatorial Guinea"[All Fields] OR ("gabon"[MeSH Terms] OR "gabon"[All Fields]) OR "Morocco"[All Fields] OR "South Sudan"[All Fields] OR "Sudan"[All Fields] OR "Botswana"[All Fields] OR "Lesotho"[All Fields] OR "Swaziland"[All Fields] OR ("eswatini"[MeSH Terms] OR "eswatini"[All Fields]) OR "Benin"[All Fields] OR "Burkina Faso"[All Fields] OR "Cape Verde"[All Fields] OR "Ghana"[All Fields] OR "Guinea"[All Fields] OR "Guinea-Bissau"[All Fields] OR "Mauritania"[All Fields] OR "Niger"[All Fields] OR "Senegal"[All Fields] OR "Sierra Leone"[All Fields] OR "Togo"[All Fields] OR ("burundi"[MeSH Terms] OR "burundi"[All Fields]) OR ("eritrea"[MeSH Terms] OR "eritrea"[All Fields]) OR ("ethiopia"[MeSH Terms] OR "ethiopia"[All Fields] OR "ethiopia s"[All Fields]) OR ("kenya"[MeSH Terms] OR "kenya"[All Fields] OR "kenya s"[All Fields]) OR ("mozambique"[MeSH Terms] OR "mozambique"[All Fields] OR "mozambique s"[All Fields]) OR ("rwanda"[MeSH Terms] OR "rwanda"[All Fields] OR "rwanda s"[All Fields]) OR ("somalia"[MeSH Terms] OR "somalia"[All Fields]) OR ("tanzania"[MeSH Terms] OR "tanzania"[All Fields] OR "tanzania s"[All Fields]) OR ("uganda"[MeSH Terms] OR "uganda"[All Fields] OR "uganda s"[All Fields]) OR ("zambia"[MeSH Terms] OR "zambia"[All Fields] OR "zambia s"[All Fields]) OR ("zimbabwe"[MeSH Terms] OR "zimbabwe"[All Fields] OR "zimbabwe s"[All Fields]) OR ("angola"[MeSH Terms] OR "angola"[All Fields] OR "angola s"[All Fields]) OR ("algeria"[MeSH Terms] OR "algeria"[All Fields]) OR ("egypt"[MeSH Terms] OR "egypt"[All Fields] OR "egypt s"[All Fields]) OR ("tunisia"[MeSH Terms] OR "tunisia"[All Fields]) OR ("namibia"[MeSH Terms] OR "namibia"[All Fields]) OR "South Africa"[All Fields] OR "the gambia"[All Fields] OR ("liberia"[MeSH Terms] OR "liberia"[All Fields] OR "liberia s"[All Fields]) OR ("mali"[MeSH Terms] OR "mali"[All Fields]) OR "Democratic Republic of Congo"[All Fields] OR "Ivory coast"[All Fields] OR ("libya"[MeSH Terms] OR "libya"[All Fields]) OR

("mauritius"[MeSH Terms] OR "mauritius"[All Fields]) OR ("nigeria"[MeSH Terms] OR "nigeria"[All Fields] OR "nigeria s"[All Fields]) OR "Sao Tome"[All Fields] OR "sub africa*"[All Fields]) AND ("Papillomaviridae"[MeSH Terms] OR ("HPV"[Title/Abstract] OR "HUMAN PAPILLOMA VIRUS"[Title/Abstract] OR "HUMAN PAPILLOMAVIRUS"[Title/Abstract])) AND "Head and Neck Neoplasms"[MeSH Terms] AND ("PREVALENCE"[Title/Abstract] OR "INCIDENCE"[Title/Abstract] OR "PRESENCE"[Title/Abstract] OR "EPIDEMIOLOGY"[Title/Abstract])

Table 2: Quality assessment tool[16]

External Validity		**points**
1	Was the study's target population a close representation of the national population in relation to relevant variables?	
2.	Was the sampling frame a true/close representation of the target population?	
3	Was some form of random selection used to select the sample?	
4	Was the likelihood of non-response bias minimal	
Total Points		
Internal Validity		
5	Was data collected directly from patients as opposed to proxy?	
6	Was an acceptable case definition used in the study?	
7	Was the study instrument that measured the parameter of interest shown to have validity and reliability?	

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8	Was the same mode of data collection used for all participants?	
9	Was the length of the shortest prevalence period for the parameter of interest appropriate?	
10	Were the numerators and denominators for the parameters of interest appropriate?	
Total points		

Table 3: Key for measurement of risk of bias[17]:

Low (0-5 points)

Further research is very unlikely to change our confidence in the estimate

Medium (6-8 points)

Further research is likely to have an important impact on our confidence in the estimate and may change the estimate

High (> 8 points)

Further research is likely

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Hoy D, Bain C, Williams G, et al. A systematic review of the global prevalence of low back pain.

17. Hoy D, Bain C, Williams G, *et al.* A systematic review of the global prevalence of low back pain. *Arthritis Rheum* 2012;**64**:2028–37. doi:10.1002/art.34347

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PRISMA-P 2015 Checklist

This checklist has been adapted for use with protocol submissions to *Systematic Reviews* from Table 3 in Moher D et al: Preferred reporting items for systematic review and meta-analysis protocols (PRISMA-P) 2015 statement. *Systematic Reviews* 2015 **4**:1

Section/topic	#	Checklist item	Information reported		Line number(s)
			Yes	No	
ADMINISTRATIVE INFORMATION					
Title					
Identification	1a	Identify the report as a protocol of a systematic review	☐		3
Update	1b	If the protocol is for an update of a previous systematic review, identify as such		☐	
Registration	2	If registered, provide the name of the registry (e.g., PROSPERO) and registration number in the Abstract	☐	☐	
Authors					
Contact	3a	Provide name, institutional affiliation, and e-mail address of all protocol authors; provide physical mailing address of corresponding author	☐		5
Contributions	3b	Describe contributions of protocol authors and identify the guarantor of the review	☐		286
Amendments	4	If the protocol represents an amendment of a previously completed or published protocol, identify as such and list changes; otherwise, state plan for documenting important protocol amendments		☐	
Support					
Sources	5a	Indicate sources of financial or other support for the review	☐		292
Sponsor	5b	Provide name for the review funder and/or sponsor		☐	
Role of sponsor/funder	5c	Describe roles of funder(s), sponsor(s), and/or institution(s), if any, in developing the protocol	☐		282
INTRODUCTION					
Rationale	6	Describe the rationale for the review in the context of what is already known	☐		100
Objectives	7	Provide an explicit statement of the question(s) the review will address with reference to participants, interventions, comparators, and outcomes (PICO)	☐		136
METHODS					

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Section/topic	#	Checklist item	Information reported		Line number(s)
			Yes	No	
Eligibility criteria	8	Specify the study characteristics (e.g., PICO, study design, setting, time frame) and report characteristics (e.g., years considered, language, publication status) to be used as criteria for eligibility for the review	☐		164,195
Information sources	9	Describe all intended information sources (e.g., electronic databases, contact with study authors, trial registers, or other grey literature sources) with planned dates of coverage	☐		196
Search strategy	10	Present draft of search strategy to be used for at least one electronic database, including planned limits, such that it could be repeated	☐		301
STUDY RECORDS					
Data management	11a	Describe the mechanism(s) that will be used to manage records and data throughout the review	☐		222
Selection process	11b	State the process that will be used for selecting studies (e.g., two independent reviewers) through each phase of the review (i.e., screening, eligibility, and inclusion in meta-analysis)	☐		215
Data collection process	11c	Describe planned method of extracting data from reports (e.g., piloting forms, done independently, in duplicate), any processes for obtaining and confirming data from investigators	☐		216, 232
Data items	12	List and define all variables for which data will be sought (e.g., PICO items, funding sources), any pre-planned data assumptions and simplifications	☐		191
Outcomes and prioritization	13	List and define all outcomes for which data will be sought, including prioritization of main and additional outcomes, with rationale	☐		169-181
Risk of bias in individual studies	14	Describe anticipated methods for assessing risk of bias of individual studies, including whether this will be done at the outcome or study level, or both; state how this information will be used in data synthesis	☐		238
DATA					
Synthesis	15a	Describe criteria under which study data will be quantitatively synthesized	☐		250
	15b	If data are appropriate for quantitative synthesis, describe planned summary measures, methods of handling data, and methods of combining data from studies, including any planned exploration of consistency (e.g., I^2, Kendall's tau)	☐		252
	15c	Describe any proposed additional analyses (e.g., sensitivity or subgroup analyses, meta-regression)	☐		257
	15d	If quantitative synthesis is not appropriate, describe the type of summary planned	☐		256
Meta-bias(es)	16	Specify any planned assessment of meta-bias(es) (e.g., publication bias across studies, selective reporting within studies)		☐	
Confidence in cumulative evidence	17	Describe how the strength of the body of evidence will be assessed (e.g., GRADE)		☐	

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